

Data Integration

□ Data integration

- Combining data from multiple sources into a coherent store

□ Schema integration: e.g., A.cust-id = B.cust-# → ชื่อต่างกัน แต่ metadata เหมือนกัน

- Integrate metadata from different sources

□ Entity identification:

- Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton

□ Detecting and resolving data value conflicts

- For the same real world entity, attribute values from different sources are different

เช่น dataset แต่ละอันอาจเก็บหน่วยเป็น ซม. / ม. / กม. / inch.

- Possible reasons: different representations, different scales, e.g., metric vs. British units

Handling Redundancy in Data Integration

- ❑ Redundant data occur often when integration of multiple databases
 - ❑ *Object identification*: The same attribute or object may have different names in different databases
 - ❑ *Derivable data*: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- ❑ Redundant attributes may be able to be detected by *correlation analysis* and *covariance analysis*
- ❑ Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Dimensionality Reduction

Modification

Reduction

Augmentation

❑ Curse of dimensionality

- ❑ When dimensionality increases, data becomes increasingly sparse
- ❑ Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
- ❑ The possible combinations of subspaces will grow exponentially

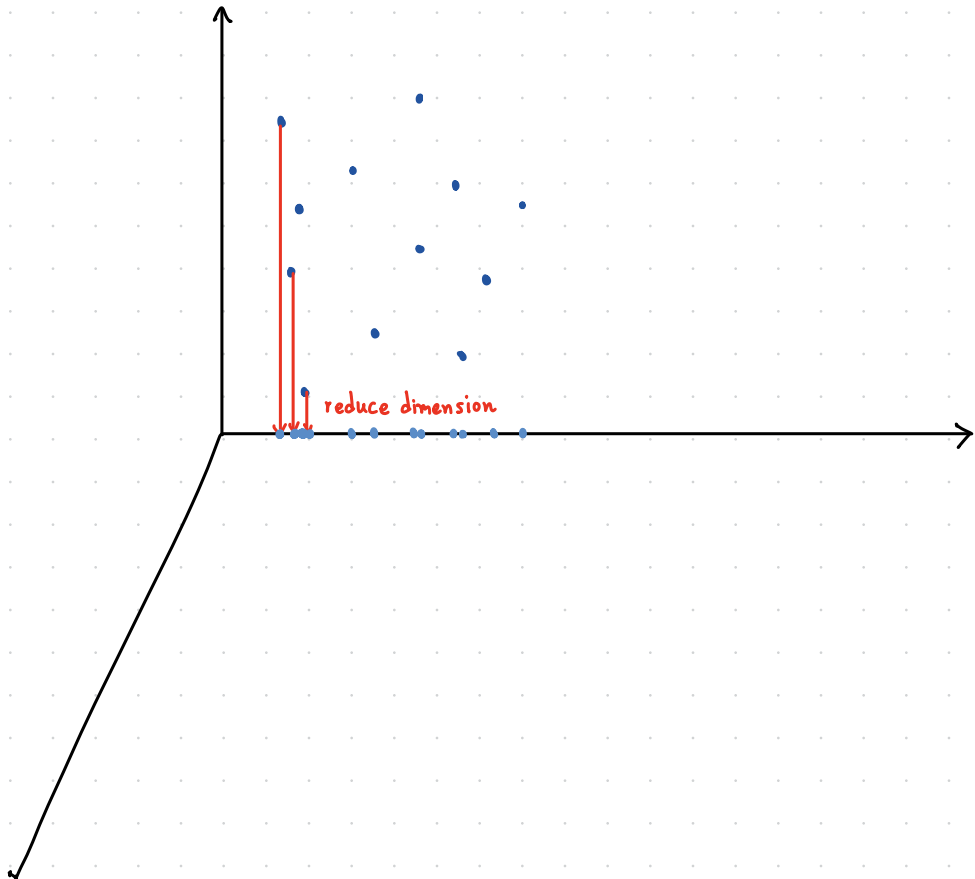
❑ Dimensionality reduction

- ❑ Reducing the number of random variables under consideration, via obtaining a set of principal variables

❑ Advantages of dimensionality reduction

- ❑ Avoid the curse of dimensionality
- ❑ Help eliminate irrelevant features and reduce noise
- ❑ Reduce time and space required in data mining
- ❑ Allow easier visualization

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Dimensionality Reduction Techniques

- Dimensionality reduction methodologies

- **Feature selection**: Find a subset of the original variables (or features, attributes)

- **Feature extraction**: Transform the data in the high-dimensional space to a space of fewer dimensions

- Some typical dimensionality methods

- **Principal Component Analysis**

- Supervised and nonlinear techniques

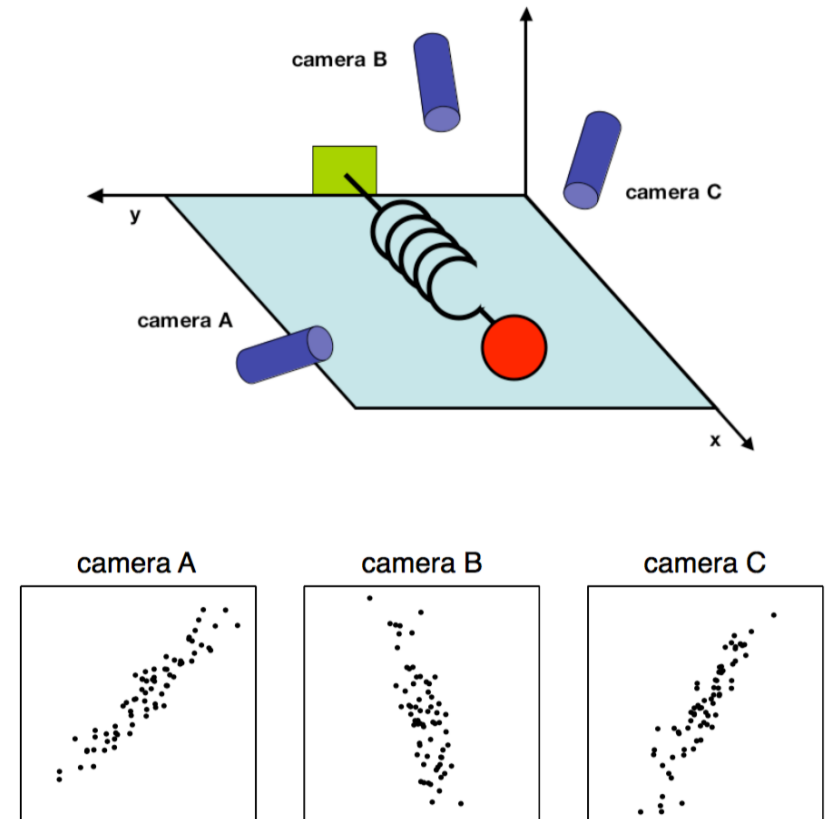
- Feature subset selection

- Feature creation

random → knowledge → q distribution → q Correlation
↓
Decision Tree

Principal Component Analysis (PCA)

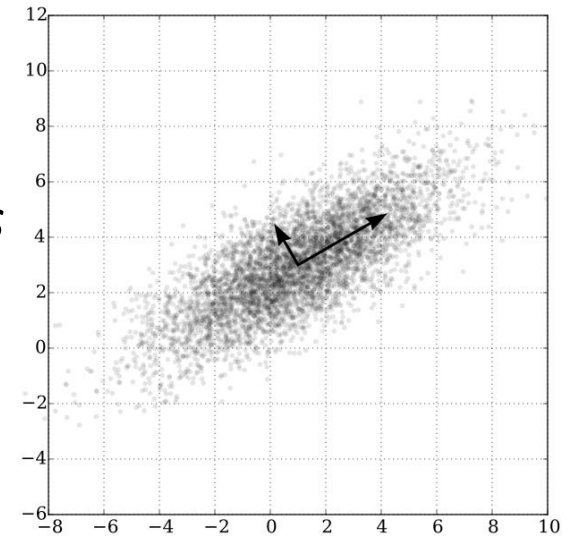
- ❑ PCA: A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called *principal components*
- ❑ The original data are projected onto a much smaller space, resulting in dimensionality reduction
- ❑ Method: Find the eigenvectors of the covariance matrix, and these eigenvectors define the new space



Ball travels in a straight line. Data from three cameras contain much redundancy

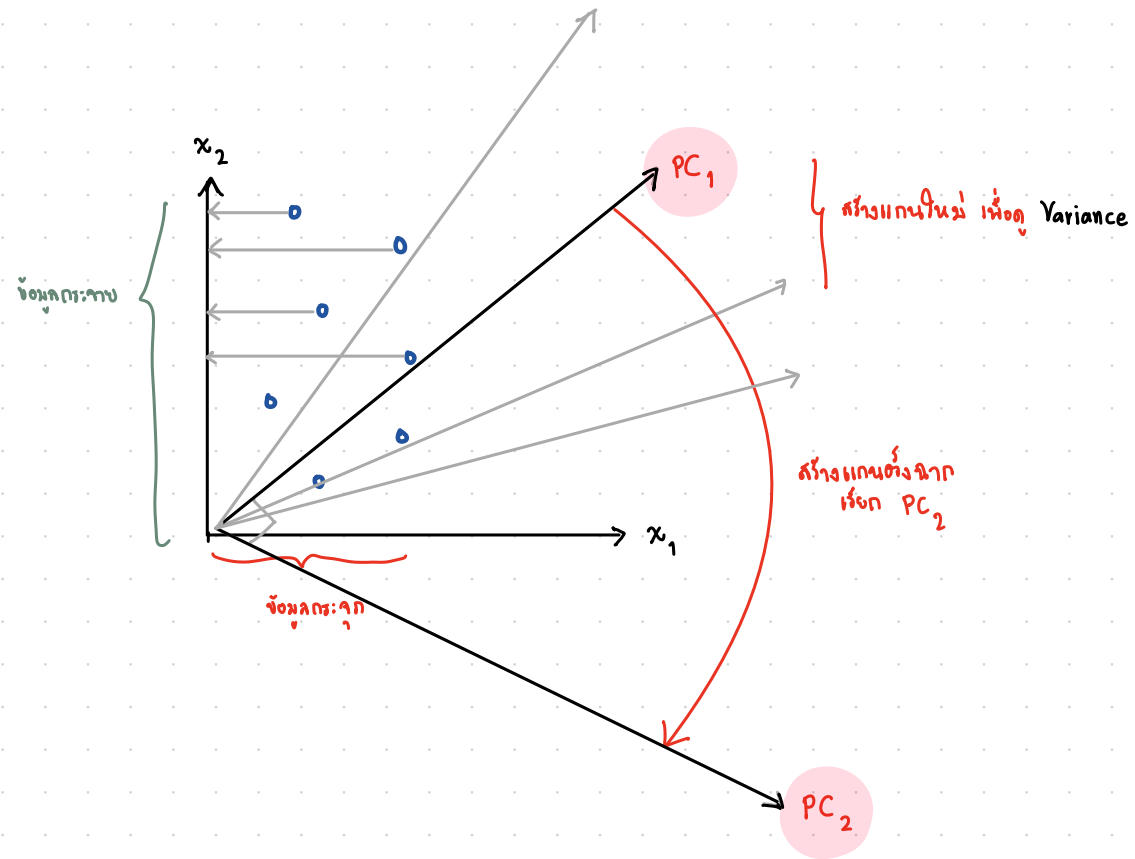
Principal Component Analysis (Method)

- Given N data vectors from n -dimensions, find $k \leq n$ orthogonal vectors (*principal components*) best used to represent data
- Normalize input data: Each attribute falls within the same range
- Compute k orthonormal (unit) vectors, i.e., *principal components*
- Each input data (vector) is a linear combination of the k principal component vectors
- The principal components are sorted in order of decreasing “significance” or strength
- Since the components are sorted, the size of the data can be reduced by eliminating the *weak components*, i.e., those with low variance (i.e., using the strongest principal components, to reconstruct a good approximation of the original data)
- Works for numeric data only



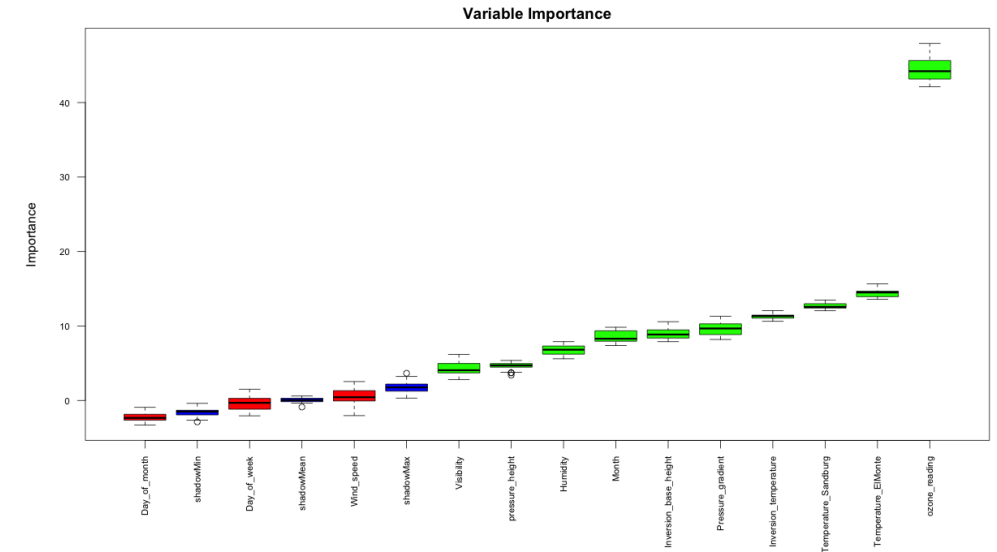
Ack. Wikipedia: Principal Component Analysis

• Principle Components Analysis



Attribute Subset Selection

- Another way to reduce dimensionality of data
- Redundant attributes
 - Duplicate much or all of the information contained in one or more other attributes
 - E.g., purchase price of a product and the amount of sales tax paid
- Irrelevant attributes
 - Contain no information that is useful for the data mining task at hand
 - Ex. A student's ID is often irrelevant to the task of predicting his/her GPA



Heuristic Search in Attribute Selection

- ❑ There are 2^d possible attribute combinations of d attributes
- ❑ Typical heuristic attribute selection methods:
 - ❑ Best single attribute under the attribute independence assumption: choose by significance tests
 - ❑ Best step-wise feature selection:
 - ❑ The best single-attribute is picked first
 - ❑ Then next best attribute condition to the first, ...
 - ❑ Step-wise attribute elimination:
 - ❑ Repeatedly eliminate the worst attribute
 - ❑ Best combined attribute selection and elimination
 - ❑ Optimal branch and bound:
 - ❑ Use attribute elimination and backtracking

Attribute Creation (Feature Generation)

- ❑ Create new attributes (features) that can capture the important information in a data set more effectively than the original ones
- ❑ Three general methodologies
 - ❑ Attribute extraction
 - ❑ Domain-specific
 - ❑ Mapping data to new space (see: data reduction)
 - ❑ E.g., Fourier transformation, wavelet transformation, manifold approaches (not covered)
 - ❑ Attribute construction
 - ❑ Combining features (see: discriminative frequent patterns in Chapter on “Advanced Classification”)
 - ❑ Data discretization

Summary

- ❑ **Data quality:** accuracy, completeness, consistency, timeliness, believability, interpretability
- ❑ **Data cleaning:** e.g. missing/noisy values, outliers
- ❑ **Data integration** from multiple sources:
 - ❑ Entity identification problem; Remove redundancies; Detect inconsistencies
- ❑ **Data reduction, data transformation and data discretization**
 - ❑ Numerosity reduction; Data compression
 - ❑ Normalization; Concept hierarchy generation
- ❑ **Dimensionality reduction**

References

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